Pharma code position may vary as per folding machine suitability.

xxx

45xxxxx
Decitabine

A. Product Name

Leudibine - Decitabine powder for concentrate for solution for infusion 50 mg/vial.

B. Name and Strength of Active Substance(s)

Decitabine 50 mg

C. Product Description

Leudibine - Decitabine powder for concentrate for solution for infusion 50 mg/vial is white to almost white lyophilized powder or cake in flint vial with rubber stopper and aluminium seal.

Description after Reconstitution:

Clear, Colorless solution

Description after further dilution:

A clear colorless solution

D. Pharmacodynamics/Pharmacokinetics

D.1. Pharmacodynamic properties

Pharmacotherapeutic group: Antineoplastic agents, antimetabolites, pyrimidine analogues.

ATC code: L01BC08.

Mechanism of action

Decitabine (5-aza-2'-deoxycytidine) is a cytidine deoxynucleoside analogue that selectively inhibits DNA methyltransferases at low doses, resulting in gene promoter hypomethylation that can result in reactivation of tumour suppressor genes, induction of cellular differentiation or cellular senescence followed by programmed cell death.

Clinical experience

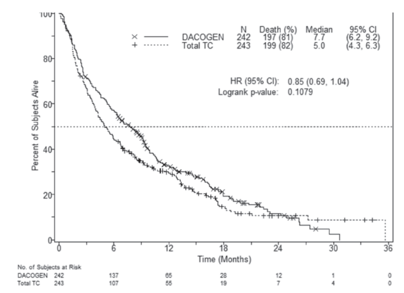
The use of Decitabine for Injection was studied in an open-label, randomised, multicentre Phase III study (DACO-016) in subjects with newly diagnosed *de novo* or secondary AML according to the WHO classification. Decitabine for Injection (n = 242) was compared to treatment choice (TC, n = 243) which consisted of patient›s choice with physician›s advice of either supportive care alone (n = 28, 11.5%) or 20 mg/m² cytarabine subcutaneously once daily for 10 consecutive days repeated every 4 weeks (n = 215, 88.5%). Decitabine for Injection was administered as a 1-hour intravenous infusion of 20 mg/m² once daily for 5 consecutive days repeated every 4 weeks.

Subjects who were considered candidates for standard induction chemotherapy were not included in the study as shown by the following baseline characteristics. The median age for the intent-to-treat (ITT) population was 73 years (range 64 to 91 years). Thirty-six percent of subjects had poor-risk cytogenetics at baseline. The remainder of the subjects had intermediate-risk cytogenetics. Patients with favourable cytogenetics were not included in the study. Twenty-five percent of subjects had an ECOG performance status ≥ 2. Eighty-one percent of subjects had significant comorbidities (e.g., infection, cardiac impairment, pulmonary impairment). The number of patients treated with Decitabine for Injection by racial group was White 209 (86.4%) and Asian 33 (13.6%).

The primary endpoint of the study was overall survival. The secondary endpoint was complete remission rate that was assessed by independent expert review. Progression-free survival and Event-free survival were tertiary endpoints.

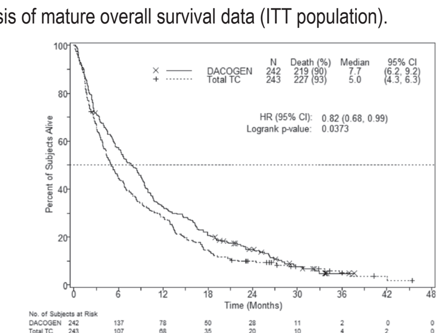
The median overall survival in the --ITT population was 7.7 months in subjects treated with Decitabine for Injection compared to 5.0 months for subjects in the TC arm (hazard ratio 0.85; 95% CI: 0.69, 1.04, p = 0.1079). The difference did not reach statistical significance, however, there was a trend for improvement in survival with a 15% reduction in the risk of death for subjects in the Decitabine for Injection arm (Figure 1). When censored for potentially disease modifying subsequent therapy (i.e., induction chemotherapy or hypomethylating agent) the analysis for overall survival showed a 20% reduction in the risk of death for subjects in the Decitabine for Injection arm [HR = 0.80, (95% CI: 0.64, 0.99), p-value = 0.0437)].

Figure 1. Overall survival (ITT population).



In an analysis with an additional 1 year of mature survival data, the effect of Decitabine for Injection on overall survival demonstrated a clinical improvement compared to the TC arm (7.7 months vs. 5.0 months, respectively, hazard ratio = 0.82, 95% CI: 0.68, 0.99, nominal p-value = 0.0373, Figure 2).

Figure 2. Analysis of mature overall survival data (ITT population).



Based on the initial analysis in the ITT population, a statistically significant difference in complete remission rate (CR + CRp) was achieved in favour of subjects in the Decitabine for Injection arm, 17.8% (43/242) compared to the TC arm, 7.8% (19/243); treatment difference 9.9% (95% CI: 4.07; 15.83), p = 0.0011. The median time to best response and median duration of best response in patients who achieved a CR or CRp were 4.3 months and 8.3 months, respectively. Progression-free survival was significantly longer for subjects in the Decitabine for Injection arm, 3.7 months (95% CI: 2.7, 4.6) compared with subjects in the TC arm, 2.1 months (95% CI: 1.9, 3.1); hazard ratio 0.75 (95% CI: 0.62, 0.91), p = 0.0031. These results as well as other endpoints are shown in Table 1.

Table 1: Other efficacy endpoints for Study DACO-016 (ITT population)

Outcomes	Decitabine n = 242	TC (combined group) n = 243	p-value
CR + CRp	43 (17.8%)	19 (7.8%)	0.0011
	OR = 2.5 (1.40, 4.78)[b]		
CR	38 (15.7%)	18 (7.4%)	-
EFS[a]	3.5 (2.5, 4.1)[b]	2.1 (1.9, 2.8)[b]	0.0025
	HR = 0.75 (0.62, 0.90)[b]		
PFS[a]	3.7 (2.7, 4.6)[b]	2.1 (1.9, 3.1)[b]	0.0031
	HR = 0.75 (0.62, 0.91)[b]		

CR = complete remission; CRp = complete remission with incomplete platelet recovery,
EFS = event-free survival, PFS = progression-free survival, OR = odds ratio, HR = hazard ratio
- = Not evaluable
[a] Reported as median months
[b] 95% confidence intervals

Overall survival and complete remission rates in pre-specified disease-related sub-groups (i.e., cytogenetic risk, Eastern Cooperative Oncology Group [ECOG] score, age, type of AML, and baseline bone marrow blast count) were consistent with results for the overall study population.

The use of Decitabine for Injection as initial therapy was also evaluated in an open-label, single-arm, Phase II study (DACO-017) in 55 subjects > 60 years with AML according to the WHO classification. The primary endpoint was complete remission (CR) rate that was assessed by independent expert review. The secondary endpoint of the study was overall survival. Decitabine for Injection was administered as a 1-hour intravenous infusion of 20 mg/m² once daily for 5 consecutive days repeated every 4 weeks. In the ITT analysis, a CR rate of 23.6% (95% CI: 13.2, 37) was observed in 13/55 subjects treated with Decitabine for Injection. The median time to CR was 4.1 months, and the median duration of CR was 18.2 months. The median overall survival in the ITT population was 7.6 months (95% CI: 5.7, 11.5).

The efficacy and safety of Decitabine for Injection has not been evaluated in patients with acute promyelocytic leukaemia or CNS leukaemia.

Paediatric population

A Phase I/II open-label, multicentre study evaluated the safety and efficacy of Decitabine for Injection in sequential administration with cytarabine in children aged 1 month to < 18 years with relapsed or refractory AML. A total of 17 subjects were enrolled and received Decitabine for Injection 20 mg/m² in this study, of which 9 subjects received cytarabine 1 g/m² and 8 subjects received cytarabine administered at the maximum tolerable dose of 2 g/m². All subjects discontinued the study treatment. The reasons for treatment discontinuation included disease progression (12 [70.6%] subjects), subjects proceeding to transplant (3 [17.6%]), investigator decision (1 [5.9%]), and “other” (1 [5.9%]). Reported adverse events were consistent with the known safety profile of Decitabine for Injection in adults. Based on these negative results, Decitabine for Injection should not be used in children with AML aged < 18 years, because efficacy was not established.

D.2. Pharmacokinetics

The population pharmacokinetic (PK) parameters of decitabine were pooled from 3 clinical studies in 45 patients with AML or myelodysplastic syndrome (MDS) utilizing the 5-Day regimen. In each study, decitabine PK was evaluated on the fifth day of the first treatment cycle.

Distribution

The pharmacokinetics of decitabine following intravenous administration as a 1-hour infusion were described by a linear two-compartment model, characterised by rapid elimination from the central compartment and by relatively slow distribution from the peripheral compartment. For a typical patient (weight 70 kg/body surface area 1.73 m²) the decitabine pharmacokinetic parameters are listed in the Table 2 below.

Table 2: Summary of population PK analysis for a typical patient receiving daily 1-hour infusions of Decitabine for Injection 20 mg/m² **over 5 days every 4 weeks**

Parameter [a]	**Predicted Value**	**95% CI**
C_{max} (ng/ml)	107	88.5 - 129
AUC_{cum} (ng.h/ml)	580	480 - 695
$t_{1/2}$ (min)	68.2	54.2 - 79.6
Vd_{ss} (L)	116	84.1 - 153
CL (L/h)	298	249 - 359

[a] The total dose per cycle was 100 mg/m²

Decitabine exhibits linear PK and following the intravenous infusion, steady-state concentrations are reached within 0.5 hour. Based on model simulation, PK parameters were independent of time (i.e., did not change from cycle to cycle) and no accumulation was observed with this dosing regimen. Plasma protein binding of decitabine is negligible (< 1%). Decitabine Vdss in cancer patients is large indicating distribution into peripheral tissues. There was no evidence of dependencies on age, creatinine clearance, total bilirubin, or disease.

Biotransformation

Intracellularly, decitabine is activated through sequential phosphorylation via phosphokinase activities to the corresponding triphosphate, which is then incorporated by the DNA polymerase. In vitro metabolism data and the human mass balance study results indicated that the cytochrome P450 system is not involved in the metabolism of decitabine. The primary route of metabolism is likely through deamination by cytidine deaminase in the liver, kidney, intestinal epithelium and blood. Results from the human mass-balance study showed that unchanged decitabine in plasma accounted for approximately 2.4% of total radioactivity in plasma. The major circulating metabolites are not believed to be pharmacologically active. The presence of these metabolites in urine together with the high total body clearance and low urinary excretion of unchanged decitabine in the urine (~4% of the dose) indicate that decitabine is appreciably metabolized in vivo. In vitro studies show that decitabine does not inhibit nor induce CYP 450 enzymes up to more than 20-fold of the therapeutic maximum observed plasma concentration (Cmax). Thus; CYP-mediated metabolic drug interactions are not anticipated, and decitabine is unlikely to interact with agents metabolized through these pathways. In addition, in vitro data show that decitabine is a poor P-gp substrate.

Elimination

Mean plasma clearance following intravenous administration in cancer subjects was > 200 L/h with moderate inter-subject variability (coefficient of variation [CV] is approximately 50%). Excretion of unchanged drug appears to play only a minor role in the elimination of decitabine.

Results from a mass balance study with radioactive 14C-decitabine in cancer patients showed that 90% of the administered dose of decitabine (4% unchanged drug) is excreted in the urine.

Additional information on special populations

The effects of renal or hepatic impairment, gender, age or race on the pharmacokinetics of decitabine have not been formally studied. Information on special populations was derived from pharmacokinetic data from the 3 studies noted above, and from one Phase I study in MDS subjects, (N = 14; 15 mg/m² x 3-hours q8h x 3 days).

Elderly

Population pharmacokinetic analysis showed that decitabine pharmacokinetics are not dependent on age (range studied 40 to 87 years; median 70 years).

Paediatric population

Population PK analysis of decitabine showed that after accounting for body size, there is no difference between decitabine PK parameters in paediatric AML patients versus adults with AML or MDS.

Gender

Population pharmacokinetic analysis of decitabine did not show any clinically relevant difference between men and women.

Race

Most of the patients studied were Caucasian. However, the population pharmacokinetic analysis of decitabine indicated that race had no apparent effect on the exposure to decitabine.

Hepatic impairment

The PK of decitabine have not been formally studied in patients with hepatic impairment. Results from a human mass-balance study and in vitro experiments mentioned above indicated that the CYP enzymes are unlikely to be involved in the metabolism of decitabine. In addition, the limited data from the population PK analysis indicated no significant PK parameter dependencies on total bilirubin concentration despite a wide range of total bilirubin levels. Thus, decitabine exposure is not likely to be affected in patients with impaired hepatic function.

Renal impairment

The PK of decitabine have not been formally studied in patients with renal insufficiency. The population PK analysis on the limited decitabine data indicated no significant PK parameter dependencies on normalized creatinine clearance, an indicator of renal function. Thus, decitabine exposure is not likely to be affected in patients with impaired renal function.

E. Indication

- Leudibine is indicated for treatment of adult patients with myelodysplastic syndromes (MDS) including previously treated and untreated, de novo and secondary MDS of all French-American-British subtypes (refractory anemia, refractory anemia with ringed sideroblasts, refractory anemia with excess blasts, refractory anemia with excess blasts in transformation, and chronic myelomonocytic leukemia) and intermediate-1, intermediate-2, and high-risk International Prognostic Scoring System groups.
- Leudibine is indicated for the treatment of adult patients aged 65 and above with newly diagnosed de novo or secondary acute myeloid leukemia (AML), according to the World Health Organization (WHO) classification, who are not candidates for standard induction chemotherapy.

F. Dosage and Administration

Leudibine administration must be initiated under the supervision of physicians experienced in the use of chemotherapeutic medicinal products.

Treatment Regimen in Myelodysplastic Syndromes (MDS)

Pre-Medications and Baseline Testing

- Consider pre-medicating for nausea with antiemetics.
- Conduct baseline laboratory testing: complete blood count (CBC) with platelets, serum hepatic panel, and serum creatinine.

Leudibine Regimen Options

Three Day Regimen

Administer Leudibine at a dose of 15 mg/m² by continuous intravenous infusion over 3 hours repeated every 8 hours for 3 days. Repeat cycles every 6 weeks upon hematologic recovery (ANC at least 1,000/µL and platelets at least 50,000//µL) for a minimum of 4 cycles. A complete or partial response may take longer than 4 cycles. Delay and reduce dose for hematologic toxicity *[see Dosage Modifications for Adverse Reaction in MDS below].*

Five Day Regimen

Administer Leudibine at a dose of 20 mg/m² by continuous infusion over 1 hour daily for 5 days. Delay and reduce dose for hematologic toxicity *[see Dosage Modifications for Adverse Reaction in MDS below]*. Repeat cycles every 4 weeks upon hematologic recovery (ANC at least 1,000/µL and platelets at least 50,000/µL) for a minimum of 4 cycles. A complete or partial response may take longer than 4 cycles.

Patients with Renal or Severe Hepatic Impairment

Treatment with Leudibine has not been studied in patients with pre-existing renal or hepatic impairment. For patients with pre-existing renal or hepatic impairment, consider the potential risks and benefits before initiating treatment with Leudibine.

Dosage Modifications for Adverse Reactions in MDS

Hematologic Toxicity

If hematologiv recovery from a previous Leudibine treatment cycle requires more than 6 weeks, delay the next cycle of Decitabine therapy and reduce Leudibine dose temporarily by following this algorithm:

- Recovery requiring more than 6, but less than 8 weeks: delay Decitabine dosing for up to 2 weeks and reduce the dose temporarily to 11 mg/m² every 8 hours (33 mg/m²/day, 99 mg/m²/cycle) upon restarting therapy.
- Recovery requiring more than 8, but less than 10 weeks: Perform bone marrow aspirate to assess for disease progression. In the absence of progression, delay Decitabine dosing for up to 2 more weeks and reduce the dose to 11 mg/m² every 8 hours (33 mg/m²/day, 99 mg/m²/cycle) upon restarting therapy, then maintain or increase dose in subsequent cycles as clinically indicated.

Non-hematologic Toxicity

Delay subsequent Leudibine treatment for any the following nonhematologic toxicities and do not restart until toxicities resolve:

- Serum creatinine greater than or equal to 2 mg/dL
- Alanine transaminase (ALT), total bilirubin greater than or equal to 2 times upper limit of normal (ULN)
- Active or uncontrolled infection

Treatment Regimen in Acute Myeloid Leukemia (AML)

In a treatment cycle, Leudibine is administered at a dose of 20 mg/m² body surface area by intravenous infusion over 1 hour repeated daily for 5 consecutive days (i.e., a total of 5 doses per treatment cycle). The total daily dose must not exceed 20 mg/m² and the total dose per treatment cycle must not exceed 100 mg/m². If a dose is missed, treatment should be resumed as soon as possible. The cycle should be repeated every 4 weeks depending on the patient’s clinical response and observed toxicity. It is recommended that patients be treated for a minimum of 4 cycles; however, a complete or partial remission may take longer than 4 cycles to be obtained. Treatment may be continued as long as the patient shows response, continues to benefit or exhibits stable disease, i.e., in the absence of overt progression.

If after 4 cycles, the patient’s hematological values (e.g., platelet count or absolute neutrophil count), have not returned to pre-treatment levels or if disease progression occurs (peripheral blast counts are increasing or bone marrow blast counts are worsening), the patient may be considered to be a non-responder and alternative therapeutic options to Leudibine should be considered.

Pre-medication for the prevention of nausea and vomiting is not routinely recommended but may be administered if required.

Management of Myelosuppression and Associated Complications In AML

Myelosuppression and adverse events related to myelosuppression (thrombocytopenia, anemia, neutropenia, and febrile neutropenia) are common in both treated and untreated patients with AML. Complications of myelosuppression include infections and bleeding. Treatment may be delayed at the discretion of the treating physician, if the patient experiences myelosuppression-associated complications, such as those described below:

- Febrile neutropenia (temperature ≥ 38.5ºC and absolute neutrophil count < 1,000/µL)
- Active viral, bacterial or fungal infection (i.e., requiring intravenous anti-infectives or extensive supportive care)
- Hemorrhage (gastrointestinal, genito-urinary, pulmonary with platelets < 25,000/µL or any central nervous system hemorrhage)

Treatment with Leudibine may be resumed once these conditions have improved or have been stabilized with adequate treatment (anti-infective therapy, transfusions, or growth factors).

In clinical studies, approximately one-third of patients receiving Decitabine required a dose-delay. Dose reduction is not recommended.

Special Populations

Pediatrics

The safety and effectiveness in pediatric patients with MDS have not been studied.

Leudibine should not be used in children with AML aged < 18 years, because efficacy was not established. Currently available data are described in sections Adverse Reactions, Clinical Studies and Pharmacokinetic Properties.

Hepatic impairment

Studies in patients with hepatic impairment have not been conducted. The need for dosage adjustment in patients with hepatic impairment has not been evaluated. If worsening hepatic function occurs, patients should be carefully monitored (see Warnings and Precautions, Pharmacokinetic Properties).

Renal impairment

Studies in patients with renal impairment have not been conducted; however, data from clinical trials that included patients with mild-moderate impairment indicated no need for dosage adjustment. Patients with severe renal impairment were excluded from these trials (see Pharmacokinetic Properties).

Leudibine is administered by intravenous infusion. A central venous catheter is not required. For instructions on reconstitution and dilution of the medicinal product before administration, see Instructions for Use and Handling and Disposal.

Size: 460(L) x 325(W) mm
Folding Size: 58(L) x 34(W) mm
Font: Times New Roman
Color: Black
Min. Font: 8pt
Market: Emerging (Malaysia)
R00 (04.09.2024)

Instructions for Use and Handling and Disposal:

This medicinal product is for single use only.

Skin contact with the solution should be avoided and protective gloves must be worn. Standard procedures for dealing with anticancer agents should be adopted.

Decitabine for Injection should be aseptically reconstituted with 10 ml of water for injections. Upon reconstitution, each ml contains approximately 5 mg of decitabine at pH 6.7 to 7.3. Immediately after reconstitution, the solution should be further diluted with 0.9% Sodium Chloride Injection or 5% Dextrose Injection to a final drug concentration of 0.15 to 1.0 mg/mL. Unless used within 15 minutes of reconstitution, the diluted solution must be prepared using cold (2°C to 8°C) infusion fluids and stored at 2°C to 8°C for up to a maximum of 4 hours until administration. Any unused product or waste material should be disposed of in accordance with local requirements.

G. Route of Administration

Intravenous Infusion

H. Contraindication

Hypersensitivity to decitabine or to any of the excipients (see List of Excipients)

Breast-feeding

I. Warning and Precautions

Myelosuppression

Myelosuppression and complications of myelosuppression, including infections and bleeding that occur in patients with AML may be exacerbated with Decitabine for Injection treatment. Therefore, patients are at increased risk for severe infections (due to any pathogen such as bacterial, fungal and viral), with potentially fatal outcome. Patients should be monitored for signs and symptoms of infection and treated promptly.

In clinical studies, the majority of patients had baseline Grade 3/4 myelosuppression. In patients with baseline Grade 2 abnormalities, worsening of myelosuppression was seen in most patients and more frequently than in patients with baseline Grade 1 or 0 abnormalities. Myelosuppression caused by Decitabine for Injection is reversible. Complete blood and platelet counts should be performed regularly, as clinically indicated and prior to each treatment cycle. In the presence of myelosuppression or its complications, treatment with Decitabine for Injection may be interrupted and/or supportive measures instituted.

Respiratory, thoracic and mediastinal disorders

Cases of interstitial lung disease (ILD) (including pulmonary infiltrates, organising pneumonia and pulmonary fibrosis) without signs of infectious aetiology have been reported in patients receiving decitabine. Careful assessment of patients with an acute onset or unexplained worsening of pulmonary symptoms should be performed to exclude ILD. If ILD is confirmed, appropriate treatment should be initiated.

Hepatic impairment

Use in patients with hepatic impairment has not been established. Caution should be exercised in the administration of Decitabine for Injection to patients with hepatic impairment and in patients who develop signs or symptoms of hepatic impairment. Liver function tests should be performed prior to initiation of therapy and prior to each treatment cycle, and as clinically indicated.

Differentiation syndrome

Cases of differentiation syndrome (also known as retinoic acid syndrome) have been reported in patients receiving decitabine. Differentiation syndrome may be fatal. Treatment with high-dose IV corticosteroids and hemodynamic monitoring should be considered at first onset of symptoms or signs suggestive of differentiation syndrome. Temporary discontinuation of Decitabine should be considered until resolution of symptoms and if resumed, caution is advised.

Renal impairment

Use in patients with severe renal impairment has not been studied. Caution should be exercised in the administration of Decitabine for Injection to patients with severe renal impairment (Creatinine Clearance [CrCl] < 30 ml/min). Renal function tests should be performed prior to initiation of therapy and prior to each treatment cycle, and as clinically indicated.

Cardiac disease

Patients with a history of severe congestive heart failure or clinically unstable cardiac disease were excluded from clinical studies and therefore, the safety and efficacy of Decitabine for Injection in these patients has not been established. Cases of cardiomyopathy with cardiac decompensation, in some cases reversible after treatment discontinuation, dose reduction or corrective treatment, have been reported in the postmarketing setting. Patients, especially those with cardiac disease history, should be monitored for signs and symptoms of heart failure.

Excipients

This medicine contains 0.5 mmol potassium per vial. After reconstitution and dilution of the solution for intravenous infusion, this medicine contains less than 1 mmol (39 mg) of potassium per dose, i.e. essentially 'potassium- free'.

This medicine contains 0.29 mmol (6.67 mg) sodium per vial. After reconstitution and dilution of the solution for intravenous infusion, this medicine contains between 13.8 mg-138 mg (0.6-6 mmol) sodium per dose (depending on the infusion fluid for dilution), equivalent to 0.7-7% of the WHO recommended maximum daily intake of 2 g sodium for an adult.

J. Interaction with Other Medicaments

No formal clinical drug interaction studies with decitabine have been conducted.

There is the potential for a drug-drug interaction with other agents which are also activated by sequential phosphorylation (via intracellular phosphokinase activities) and/or metabolised by enzymes implicated in the inactivation of decitabine (e.g., cytidine deaminase). Therefore, caution should be exercised if these active substances are combined with decitabine.

Impact of co-administered medicinal products on decitabine

Cytochrome (CYP) 450-mediated metabolic interactions are not anticipated as decitabine metabolism is not mediated by this system but by oxidative deamination.

Impact of decitabine on co-administered medicinal products

Given its low *in vitro* plasma protein binding (< 1%), decitabine is unlikely to displace co-administered medicinal products from their plasma protein binding. Decitabine has been shown to be a weak inhibitor of P-gp mediated transport *in vitro* and is therefore, also not expected to affect P-gp mediated transport of co-administered medicinal products.

K. Statement on usage during pregnancy and lactation

Women of childbearing potential/Contraception in men and women

Due to the genotoxic potential of decitabine, women of childbearing potential must use effective contraceptive measures and avoid becoming pregnant while being treated with Decitabine for Injection and for 6 months following completion of treatment. Men should use effective contraceptive measures and be advised to not father a child while receiving Decitabine for Injection., and for 3 months following completion of treatment. The use of decitabine with hormonal contraceptives has not been studied.

Pregnancy

There are no adequate data on the use of Decitabine for Injection in pregnant women. Studies have shown that decitabine is teratogenic in rats and mice. The potential risk for humans is unknown. Based on results from animal studies and its mechanism of action, Decitabine for Injection should not be used during pregnancy and in women of childbearing potential not using effective contraception. A pregnancy test should be performed on all women of childbearing potential before treatment is started. If Decitabine for Injection is used during pregnancy, or if a patient becomes pregnant while receiving this medicinal product, the patient should be apprised of the potential hazard to the foetus.

Breast-feeding

It is not known whether decitabine or its metabolites are excreted in breast milk. Decitabine for Injection is contraindicated during breast-feeding; therefore, if treatment with this medicine is required, breast-feeding must be discontinued.

Fertility

No human data on the effect of decitabine on fertility are available. In non-clinical animal studies, decitabine alters male fertility and is mutagenic. Because of the possibility of infertility as a consequence of Decitabine for Injection therapy, men should seek advice on conservation of sperm and female patients of childbearing potential should seek consultation regarding oocyte cryopreservation prior to initiation of treatment.

L. Adverse Effects/ Undesirable Effects

Summary of the safety profile

The most common adverse drug reactions (≥ 35%) reported are pyrexia, anaemia and thrombocytopaenia.

The most common Grade 3/4 adverse drug reactions (≥ 20%) included pneumonia, thrombocytopaenia, neutropaenia, febrile neutropaenia and anaemia.

In clinical studies, 30% of patients treated with Decitabine for Injection and 25% of patients treated in the comparator arm had adverse events with an outcome of death during treatment or within 30 days after the last dose of study drug.

In the Decitabine for Injection treatment group, there was a higher incidence of treatment discontinuation due to adverse events in women compared to men (43% versus 32%).

Tabulated list of adverse drug reactions

Adverse drug reactions reported in 293 AML patients treated with Decitabine for Injection are summarised in Table 3. The following table reflects data from AML clinical studies and from post-marketing experience. The adverse drug reactions are listed by frequency category. Frequency categories are defined as follows: Very common (≥ 1/10), common (≥ 1/100 to < 1/10), uncommon (≥ 1/1,000 to < 1/100), rare (≥ 1/10,000 to < 1/1,000), very rare (< 1/10,000), not known (frequency cannot be estimated from the available data).

Within each frequency grouping, adverse drug reactions are presented in order of decreasing seriousness.

Table 3: Adverse drug reactions identified with Decitabine for Injection

System Organ Class	Frequency (all Grades)	Adverse Drug Reaction	Frequency	
			All Grades[a] (%)	Grades 3-4[a] (%)
Infections and infestations	Very common	pneumonia*	24	20
		urinary tract infection*	15	7
		All other infections (viral, bacterial, fungal)*, [b, c, d]	63	39
	Common	septic shock*	6	4
		sepsis*	9	8
		sinusitis	3	1
Blood and lymphatic disorders	Very common	febrile neutropaenia*	34	32
		neutropaenia*	32	30
		thrombocytopaenia*, [e]	41	38
		anaemia	38	31
		leukopaenia	20	18
	Uncommon	pancytopaenia*	< 1	< 1
Immune system disorders	Common	hypersensitivity including anaphylactic reaction[f]	1	< 1
Metabolism and nutrition disorders	Very common	hyperglycaemia	13	3
Nervous system disorders	Very common	headache	16	1
Cardiac disorders	Uncommon	Cardiomyopathy	< 1	< 1
Respiratory, thoracic and mediastinal disorders	Very common	epistaxis	14	2
	Not known	interstitial lung disease	Not known	Not known
Gastrointestinal disorders	Very common	diarrhoea	31	2
		vomiting	18	1
		nausea	33	< 1
	Common	stomatitis	7	1
	Not known	enterocolitis, including neutropaenic colitis, caecitis*	Not known	Not known
Hepatobiliary disorders	Very common	hepatic function abnormal	11	3
	Common	hyperbilirubinaemia[g]	5	<1
Skin and subcutaneous tissue disorders	Uncommon	acute febrile neutrophilic dermatosis (Sweet's syndrome)	< 1	NA
General disorders and administration site conditions	Very common	pyrexia	48	9

[a] Worst National Cancer Institute Common Terminology Criteria for Adverse Events Grade.

[b] Excluding pneumonia, urinary tract infection, sepsis, septic shock and sinusitis.

[c] The most frequently reported "other infections" in study DACO-016 were: oral herpes, oral candidiasis, pharyngitis, upper respiratory tract infection, cellulitis, bronchitis, nasopharyngitis.

[d] Including enterocolitis infectious.

[e] Including haemorrhage associated with thrombocytopaenia, including fatal cases.

[f] Including preferred terms hypersensitivity, drug hypersensitivity, anaphylactic reaction, anaphylactic shock, anaphylactoid reaction, anaphylactoid shock.

[g] In clinical studies in AML and myelodysplastic syndrome (MDS), the reporting frequency for hyperbilirubinaemia was 11% for All Grades and 2% for Grade 3-4.

* Includes events with a fatal outcome.

NA = Not applicable

Description of selected adverse drug reactions

Haematologic adverse drug reactions

The most commonly reported haematologic adverse drug reactions associated with Decitabine for Injection treatment included febrile neutropaenia, thrombocytopaenia, neutropaenia, anaemia and leukopaenia.

Serious bleeding-related adverse drug reactions, some of which lead to a fatal outcome, such as central nervous system (CNS) haemorrhage (2%) and gastrointestinal (GI) haemorrhage (2%), in the context of severe thrombocytopaenia, were reported in patients receiving Decitabine.

Haematological adverse drug reactions should be managed by routine monitoring of complete blood counts and early administration of supportive treatments as required. Supportive treatments include, administration of prophylactic antibiotics and/or growth factor support (e.g., G-CSF) for neutropaenia and transfusions for anaemia or thrombocytopaenia according to institutional guidelines. For situations where decitabine administration should be delayed.

Infections and infestations adverse drug reactions

Serious infection-related adverse drug reactions, with potentially fatal outcome, such as septic shock, sepsis, pneumonia, and other infections (viral, bacterial and fungal) were reported in patients receiving Decitabine.

Gastrointestinal disorders

Occurrences of enterocolitis, including neutropaenic colitis, caecitis have been reported during treatment with decitabine. Enterocolitis may lead to septic complications and may be associated with fatal outcome.

Respiratory, thoracic and mediastinal disorders

Cases of interstitial lung disease (including pulmonary infiltrates, organising pneumonia and pulmonary fibrosis) without signs of infectious aetiology have been reported in patients receiving Decitabine.

Paediatric population

The safety assessment in paediatric patients is based on the limited safety data from a Phase I/II study to evaluate pharmacokinetics, safety and efficacy of Decitabine for Injection in paediatric patients (aged 1 to 14 years) with relapsed or refractory AML (n = 17). No new safety signal was observed in this paediatric study.

Adverse Effects/ Undesirable Effects:

Neoplasms benign, malignant and unspecified (including cysts and polyps)

Frequency 'Very rare': Differentiation syndrome

M. Overdose and Treatment

There is no direct experience of human overdose and no specific antidote. However, early clinical study data in published literature at doses greater than 20 times higher than the current therapeutic dose, reported increased myelosuppression including prolonged neutropaenia and thrombocytopaenia. Toxicity is likely to manifest as exacerbations of adverse drug reactions, primarily myelosuppression. Treatment for overdose should be supportive.

N. Effect on Ability to Drive and Use Machine

Decitabine for Injection has moderate influence on the ability to drive and use machines. Patients should be advised that they may experience undesirable effects such as anaemia during treatment. Therefore, caution should be recommended when driving a car or operating machines.

O. Incompatibilities:

Decitabine for Injection must be diluted prior to use with either 0.9% Sodium Chloride Injection or 5% Dextrose Injection, Decitabine for Injection should not be mixed with other medicinal products.

P. Storage Condition

Unopened vial: Store below 30°C
After reconstitution/ dilution: Unless used within 15 minutes of reconstitution, the diluted solution must be prepared using cold (2°C to 8°C) infusion fluids and stored at 2°C to 8°C for up to a maximum of 4 hours until administration.

Q. Dosage Forms and packaging available

White to almost white lyophilized powder or cake filled in 20 mL/ 20 mm flint tubular flat bottom type-I clear colorless glass vial with reinforced Non-PVC base, stoppered with 20 mm dark grey chlorobutyl flurotec coated single slotted rubber stopper and sealed with 20 mm aluminium flip-off light blue colour seal.

R. Name and Address of manufacturer/ Product registration holder

Manufactured by:
Shilpa Medicare Limited,
Unit IV, Plot S-20 To S-26, Pharma SEZ,
TSIIC, Green Industrial Park, Polepally (V)
Jadcherla (Md), Mahabubnagar (Dt.),
Telangana - 509301, India.

Product Registration Holder /Imported by:
Unimed Sdn. Bhd.,
53, Jalan Tembaga SD5/2B, Bandar
Sri Damansara,52200 Kuala Lumpur, Malaysia.

S. Date of Revision of Package Insert

September - 2024

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